

Program-5 Continuous Integration using Jenkins with Docker

To install and run Jenkins using Docker, configure Jenkins, and create a basic Continuous Integration (CI) pipeline.

2. Software Requirements

Component	Requirement
OS	Ubuntu / Linux
Docker	20.x or later
Docker Compose	1.29+
Git	Latest
Jenkins	LTS (via Docker image)
Browser	Chrome / Firefox

3. Folder Structure

```
program7/  
├── docker-compose.yml  
└── README.md
```

4. docker-compose.yml

```
version: "3.9"  
  
services:  
  
  jenkins:  
  
    image: jenkins/jenkins:lts  
    container_name: jenkins-ci  
  
    ports:  
  
      - "8080:8080"  
      - "50000:50000"  
  
    volumes:  
  
      - jenkins_home:/var/jenkins_home  
  
volumes:  
  
  jenkins_home:
```

Step 2: Start Jenkins

```
sudo docker-compose up -d
```

```
Jan 2 14:00
trv24mc064_manasas@trv24mc064manasa: ~/programs

=> sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4ea 189B / 189B 6.0s
=> sha256:687044b042c5d657e2ce207c00ee159f2382cb84c8d80688 95.32MB / 95.32MB 54.3s
=> sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342 183B / 183B 0.7s
=> sha256:68cdebf48099293f6a12f5c50b91cf4e1940b6293334d8a31a4501 1.23KB / 1.23KB 0.9s
=> sha256:2f1ecb09fa362eb652038ecbfcc64a04b042591e94ed6b0059a41 5.24MB / 5.24MB 4.2s
=> sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c 65.88MB / 65.88MB 31.0s
=> sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc00d612dc14fe36 49.29MB / 49.29MB 33.9s
=> extracting sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc00d612dc14fe368753e7be 1.5s
=> extracting sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce3 1.9s
=> extracting sha256:2f1ecb09fa362eb652038ecbfcc64a04b042591e94ed6b0059a4127c01 0.1s
=> extracting sha256:68cdebf48099293f6a12f5c50b91cf4e1940b6293334d8a31a451ed63c 0.0s
=> extracting sha256:e0e73f34dee93322ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c 0.0s
=> extracting sha256:687044b042c5d657e2ce207c00ee159f2382cb84c8d80688c35e7f1e 0.3s
=> extracting sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4ea 0.0s
=> extracting sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2df88e9c009f6874e7221ac 0.1s
=> extracting sha256:d7dd2e63023897146ec4b913a460382968fbd1cc3682e36f650727a1bf 0.9s
=> extracting sha256:3fb70446889ebcc58e91f43f90aafa314798941c318ce016ea62068a3de 0.0s
=> extracting sha256:986be6969b0a04975a2e559b30a96d766bf360d0b340b2789b611688ba8 0.0s
=> extracting sha256:d2519f76d4c6527e0b08472742da3ae15b0c5897d464be4dd93de4f5737 0.0s
[2/2] RUN apt-get update && apt-get install -y docker.io 61.4s
=> exporting to image 31.0s
=> exporting layers 27.0s
=> exporting manifest sha256:f1a6a38ec42bc1a16e62ab493f6db6dd21b9bf79334d7159042 0.0s
=> exporting config sha256:ca11205c1f8eca981fe477f8254622fb946bc0b0ade3b00c31d54 0.0s
=> exporting attestation manifest sha256:bed0b6b36875c2f3fdd8f1dd668d14d59da050 0.0s
=> exporting manifest list sha256:4320b09c3d99db4425def0cd0a9cfe9c26d7f73c6721e 0.0s
=> naming to docker.io/library/program5_jenkins:latest 0.0s
=> unpacking to docker.io/library/program5_jenkins:latest 3.9s
Creating jenkins-ci ... done
trv24mc064_manasas@trv24mc064manasa:~/programs$ docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
Error response from daemon: No such container: jenkins
trv24mc064_manasas@trv24mc064manasa:~/programs$ docker ps
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
trv24mc064_manasas@trv24mc064manasa:~/programs$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
2ec6c495848d   program5_jenkins                    "/usr/bin/tini -- /u-"   About a minute ago   Up About a minute   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->5
8000/tcp
jenkins-ci
trv24mc064_manasas@trv24mc064manasa:~/programs$ sudo docker exec jenkins-ci cat /var/jenkins_home/secrets/initialAdminPassword
Iad966f6c0b041c59c0a0e77e4802528
trv24mc064_manasas@trv24mc064manasa:~/programs$ sudo docker stop jenkins-ci
jenkins-ci
trv24mc064_manasas@trv24mc064manasa:~/programs$ sudo docker-compose up -d
Starting jenkins-ci ... done
trv24mc064_manasas@trv24mc064manasa:~/programs$
```

Step 3: Verify Jenkins Container

docker ps

Step 4: Open Jenkins in Browser

http://localhost:8080

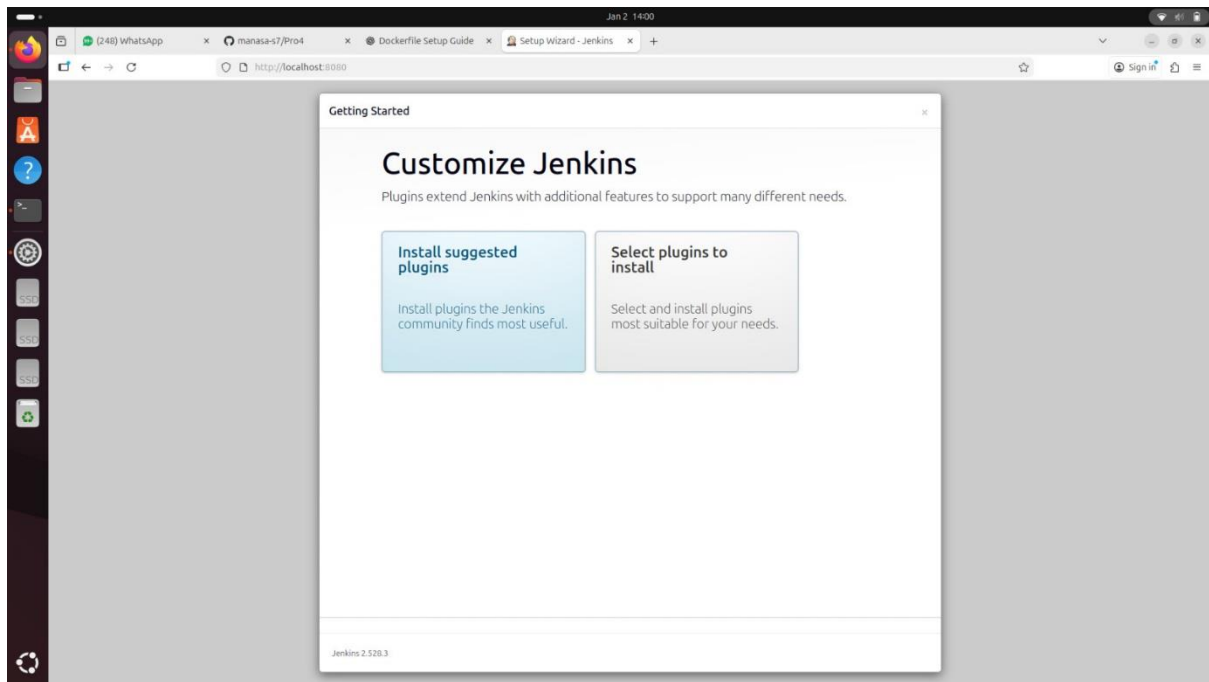
Step 5: Get Initial Admin Password

sudo docker exec jenkins-ci cat /var/jenkins_home/secrets/initialAdminPassword

Paste it in the browser.

6. Output

- Jenkins server running on browser
- Pipeline executed successfully
- Build, test, and deploy stages completed



7. Result

Jenkins was successfully installed using Docker and a basic CI pipeline was created and executed.

8. Conclusion

This experiment demonstrated the use of Jenkins for continuous integration by automating build, test, and deployment steps using a pipeline.